RAJALAKSHMI ENGINEERING COLLEGE

TAMILSELVAN P 251801349-2025-AIDS T2

Started on	Tuesday, 25 August 2026, 11:54 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:55 PM
Time taken	36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

TAMILSELVAN P 251801349-2025-AIDS

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x) {
    if (low > high)
        return -1; // If no element is <= x

    // If last element is smaller than or equal to x, it's the floor
    if (x >= arr[high])
        return arr[high];

    int mid = low + (high - low) / 2;

    // If mid element is equal to x, floor is x
    if (arr[mid] == x)
        return arr[mid];

    // If x lies between mid-1 and mid, then arr[mid-1] is the floor
    if (mid > 0 && arr[mid - 1] <= x && x < arr[mid])
        return arr[mid - 1];

    // If x is smaller than mid, search in left half
    if (x < arr[mid])
        return findFloor(arr, low, mid - 1, x);

    // If x is greater than mid, search in right half
    return findFloor(arr, mid + 1, high, x);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int floorVal = findFloor(arr, 0, n - 1, x);
    printf("%d\n", floorVal);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

TAMILSELVAN P 251801349-2025-AIDS

Back to Course